Ray Sun

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EDUCATION

University of Southern California **May 2026**
Ph.D., Electrical Engineering (M.S. December 2023), GPA: 4.0 *Los Angeles, CA*

California Institute of Technology **June 2020**
B.S., Electrical Engineering *Pasadena, CA*

RESEARCH EXPERIENCE

Graduate Researcher **August 2020 – Present**
Analog/RF IC, Microsystems, and Electromagnetics Lab, University of Southern California *Los Angeles, CA*

- Designed the first-ever integrated pulsed electron paramagnetic resonance spectrometer (14GHz, 65nm CMOS) enabling miniaturized point-of-care biomedical and diagnostics applications. Presented at ISSCC 2024.
- Designed concurrent dual-frequency drift-compensated magnetic spectrometer (1.2-1.7 / 2.9-4 GHz, 65nm CMOS) enabling single-step and wash-free magnetic immunoassays on-chip. Presented at ESSCIRC 2022.
- Designed Xilinx FPGA-based data acquisition systems, RF PCBs.

Undergraduate Research Assistant **June 2017 – March 2018**
Aerospace Robotics and Control Lab, Caltech *Pasadena, CA*

- Development of spacecraft simulators: designed STM32-based thruster controller boards, hardware selection.

WORK EXPERIENCE

Micro-Vu Corp. **June 2019 – August 2019**
Electrical Engineering Intern *Windsor, CA*

- Supported hardware and Xilinx FPGA development for non-contact and multi-sensor metrology machines.
- Developed prototype low-latency, fault-robust Bluetooth machine controller with STM32, SiLabs Blue Gecko.

Ampaire, Inc. **August 2018 – September 2018**
Powertrain Intern *Los Angeles, CA*

- Assembled and validated high voltage electric aircraft powertrain modules for ground testbed and flight aircraft.
- Developed and validated 15 Mbps isolated dual-channel CAN transceiver board.

VOLUNTEER EXPERIENCE

- Mentor to USC Solar Car team and USC Makers team, USC engineering graduate student association senator.
- Caltech Formula SAE Electric: Designed 2 generations of STM32 vehicle input PCB

SKILLS

Integrated Circuits: Cadence Virtuoso, Ansys HFSS, Cadence EMX, TSMC PDK
Hardware/Firmware: Altium Designer, KiCad, LTSpice, STM32, ARM, Xilinx FPGA, AVR, spectrum analyzer, VNA
Programming: C/C++, Python, Verilog, Assembly (AVR/ARM)
Miscellaneous: Amateur radio license (General class), Mandarin Chinese (spoken), German (elementary)

PROJECTS

Jackrabbit: Logic probe, 100 MHz bandwidth, 16-channel, RP2040
SAR ADC: 8-bit time-interleaved, 45nm (schematic), 1 GS/s, 11 mW, 5.5 ENOB
OTA: 2-stage DC-coupled, 45nm (schematic); 2.2 GHz unity-gain bandwidth, 11 mA, rail-rail input